

Adding Integers: Financial Reasoning

Adding Integers Using Financial Reasoning

Number lines and the Orange-Blue Rule are great for visualizing addition and finding answers to simple problems, but they aren't always practical. For example, we would need a very large number line to find the value of $-185 + 99$. Instead, we can add integers by thinking in terms of money.

- **Assets (+)** include money that you **have**, a gift you receive or a deposit.
- **Debts (-)** include money that you **owe**, a bill you pay or a withdrawal.

When we add integers that represent assets and debts, we are finding the **net balance**.

$$\text{Net Balance} = \text{Assets} + \text{Debts}$$

Remark 1. A statement such as "a debt of \$50" is represented mathematically as -50 . The negative sign is omitted in the phrase because the word *debt* implies a negative value.

Example 1. Combining Assets

If a child has \$100 (+100) in her piggy bank and receives \$30(+30) for doing her chores, her net balance is **\$130**.

$$100 + 30 = 130$$

Example 2. Combining Debts

If you have a credit card debt of \$500 (-500) and make another credit card purchase of \$100 (-100), your net balance is a **debt** of **\$600** (-600).

$$-500 + (-100) = -600$$

Example 3. Partial Cancellation

If you have \$20 (+20) and owe a friend \$25 (-25). Your \$20 isn't enough to cover the entire debt, it only cancels **part** of it. After you pay your friend \$20, you will still owe \$5, so your net balance is a **debt** of **\$5** (-5).

$$20 + (-25) = -5$$

Example 4. Full Cancellation

If a taxpayer **owes** the IRS \$300 (-300) and has \$800 ($+800$) in the bank, they can pay the entire debt and keep the change. Their net balance is **\$500** ($+500$).

$$-300 + 800 = 500$$

Key Observations

If you take a close look at the examples above you will notice the following relationships between the signs of the numbers that were added and the operation we actually performed.

Same Signs: When we add two assets or two debts, we **add** their absolute values. The sign of the answer is the same as the sign of the numbers.

Different Signs: When we add an asset and a debt, all or part of the debt will be removed or cancelled, so even though we are technically *adding* integers, the arithmetic performed is **subtraction**. The sign of the answer depends on which is greater: the asset or the debt.

Problem 1 Your checking account is overdrawn by \$50 and the bank charges you a \$15 overdraft fee. Write the addition problem to find your net balance. Then calculate your net balance. **If the net balance is a debt, use a negative sign in your answer.**

Multiple Choice:

- (a) $50 + (-15)$
- (b) $-50 + 15$
- (c) $50 + 15$
- (d) $-50 + (-15)$ ✓

Net Balance = -65

Problem 2 A soccer team spent \$25 on supplies for a car wash. By the end of the day, they earned \$175. Write the addition problem to find the team's net balance. Then calculate their net balance. **If the net balance is a debt, use a negative sign in your answer.**

Multiple Choice:

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(a) $25 + 175$

(b) $-25 + 175$ ✓

(c) $-25 + (-175)$

(d) $25 + (-175)$

Net Balance = 150

Problem 3 *It's your birthday! You receive \$300 as a gift, but you owe your roommate \$500 for rent. Write the addition problem to find your net balance after you use your gift money toward the rent. Then calculate your net balance. If the net balance is a debt, use a negative sign in your answer.*

Multiple Choice:

(a) $-300 + (-500)$

(b) $300 + 500$

(c) $300 + (-500)$ ✓

(d) $-300 + 500$

Net Balance = -200

Problem 4 *Think in terms of assets and debts to find the following sums.*

$$80 + (-2) = 78$$

$$-9 + (-2) = -11$$

$$-30 + 27 = -3$$